

Optics Letters

Polarization-insensitive broadband 2×2 3 dB power splitter based on silicon-bent directional couplers

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Received 4 August 2017; revised 24 August 2017; accepted 24 August 2017; posted 29 August 2017 (Doc. ID 303722); published 19 September 2017

We present and demonstrate a broadband and polarization-insensitive 2×2 3 dB power splitters based on silicon-bent directional couplers (DCs). The working bandwidth can be from 1520 to 1630 nm for both TE and TM polarizations, with an excess loss less than 1 dB. The total size for the fabricated device can be as small as 50 μm , and the measurement results show good agreement with the simulations. © 2017 Optical Society of America

OCIS codes: (130.3120) Integrated optics devices; (130.5440) Polarization-selective devices; (230.0230) Optical devices.

<https://doi.org/10.1364/OL.42.003738>

An optical power splitter is one of the key building blocks for splitting and combining optical signals in photonic integrated circuits. Silicon-based devices have shown an irreplaceable position in integrated optics due to their high refractive index contrast with silicon dioxide and the compatibility with the traditional complementary metal oxide semiconductor (CMOS) fabrication technology. A compact, low-loss, broadband and polarization-insensitive silicon-based power coupler is highly required for optical switches [1–3] and multiplexers [4,5]. A directional coupler (DC) is one of the most popular methods to realize power splitters. However, the performances of the power splitters based on conventional silicon DCs are highly sensitive to the wavelength and polarization. There have been numerous designs demonstrated to enhancing the bandwidth of the DCs, including asymmetric DCs [6], bent DCs [2,7], sub-wavelength grating structures [8,9], and hybrid plasmonic waveguides [10]. However, few works have reported on how to achieve polarization-insensitive power splitters.

In order to achieve a broadband and polarization-insensitive 3 dB power splitter, we proposed and demonstrated a novel structure based on cascaded bent DCs. In this Letter, we solve the polarization-dependent issue by utilizing two cascaded bent DCs.

The schematic diagram of the proposed 2×2 3 dB power splitter is shown in Fig. 1. The structure mainly consists of two cascaded bent DCs, i.e., DC#1 and DC#2. A straight

waveguide and an S-bend waveguide are used to connect the two outputs of DC#1 with the two inputs of DC#2. The S-bend waveguide is used so that the gap of the DC#2 can be redefined. The devices are designed based on the silicon-on-insulator (SOI) platform with the thickness of 220 nm for the silicon layer and 2 μm for the buried silica layer. In order to simplify the design, the widths of all the waveguides are set to be the same ($W_1 = W_2 = 500$ nm).

The first bent DC (DC#1) is designed to split the lights of TE polarizations at 50%:50%. For DC#1, the splitting ratio for the light of TM polarization will not be precisely controlled. Since DC#2 is designed to have a weak coupling for TE polarization, the light of TE polarization will pass through DC#2 while keeping the splitting ratio of 50%:50%. By optimizing the gaps and bending radius for DC#2, the light of TM polarization can be split with a ratio of 50%:50%. There is no need to know the exact TM split ratio for the first or the second DC. The whole 3 dB coupler can work for both polarizations with two cascaded bent DCs. Bent DCs are utilized instead of the conventional DC, since we can have more parameters to optimize, and the working bandwidth can be large.

To analyze the power transfer process in the bent DCs, the effective indices for the TE and TM supermodes are calculated with a finite element method. Figure 2(a) shows the calculated effective index differences for the TE and TM supermodes for the bent DC with gap = 300 nm and R_{central} ($R_{\text{central}} = (R_{\text{inner}} + R_{\text{outer}})/2$) from 5 to 100 μm . In order to obtain a weak coupling for TE polarization, the overlap integral [11] between the TE_{outer} supermode and TE_0 mode in the single inner waveguide should be small enough [12]. For example, when $R_{\text{central}} = 20$ μm , the effective index difference

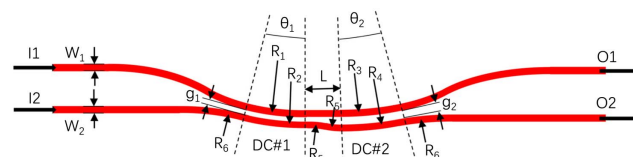


Fig. 1. Schematic configuration of the proposed 2×2 3 dB power splitter with bent DCs.

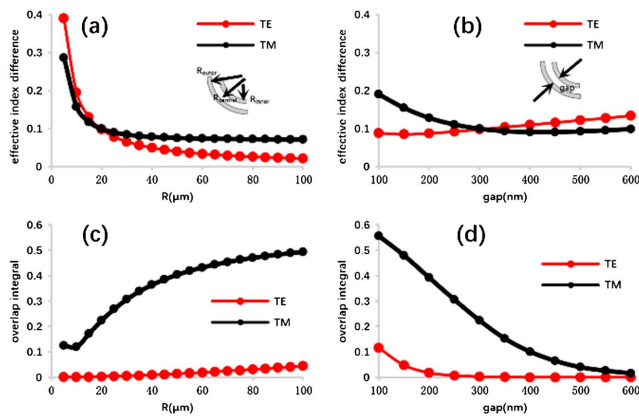


Fig. 2. (a) Effective index differences for the TE and TM supermodes with different R_{central} . (b) Effective index differences for the TE and TM supermodes with different gaps. (c) Overlap integral for the TE and TM with different R_{central} . (d) Overlap integral for the TE and TM with different gaps.

for the TE supermode is ~ 0.1 , and the overlap integral is ~ 0.0028 , which means a weak coupling for TE polarization. Weak cross-coupling for TE polarization can be obtained if R_{central} is less than $20 \mu\text{m}$. One may also notice that the effective index difference of TM polarization is as large as that of TE from Fig. 2(a). However, the cross-coupling strength of TM polarization is much stronger than TE polarization, since the mode confinement is much weaker. Figure 2(c) shows the overlap integral for the TE and TM modes with gap = 300 nm and R_{central} from 5 to $100 \mu\text{m}$. The overlap integral between the TM_{outer} supermode and TM_0 mode is about 0.224 when $R_{\text{central}} = 20 \mu\text{m}$, which means much stronger cross-coupling than TE. The gap between the two bent waveguides also has a great impact on the overlap integral. From Fig. 2(b), one can easily find that the refractive index difference of the TE supermode will increase with the gap width. However, from Fig. 2(d), we can clearly find that if the gap is too large, the coupling for TM polarization will also be too weak. Finally, we choose the gap $g_1 = 110 \text{ nm}$, the bending radius $R_1 = 35 \mu\text{m}$ ($R_2 = R_1 + W + g_1$) for DC#1 and $g_2 = 350 \text{ nm}$, and the bending radius $R_3 = 20 \mu\text{m}$ ($R_4 = R_3 + W + g_2$) for DC#2.

The three-dimensional finite-difference time-domain method is used to simulate and optimize the structure. Figure 3(a) shows the calculated transmittance of the TE polarization with different angle lengths θ_1 when the light is injected into Port 1. The angle length $\theta_1 = 18^\circ$ is chosen in order to obtain a maximal bandwidth. The designed bent DC has a coupling ratio varying from $45\%:55\%$ to $50\%:50\%$ in the wavelength band from 1490 to 1630 nm for TE polarization, which means a broad working bandwidth ($\sim 140 \text{ nm}$). The 3 dB coupler also works when light is injected into Port 2, showing good 2×2 performance. Figure 3(b) shows the calculated transmittance of the TM mode with different angle lengths θ_2 . From the calculation results, the angle length is chosen to be $\theta_2 = 20^\circ$, and the working bandwidth can be from 1520 to 1650 nm . One can find that the DC#2 has negligible influence on the transmission for TE polarization as we expected. Since g_2 is larger than g_1 , a straight waveguide and an S-bend waveguide is utilized to connect the two bent DCs. We choose the

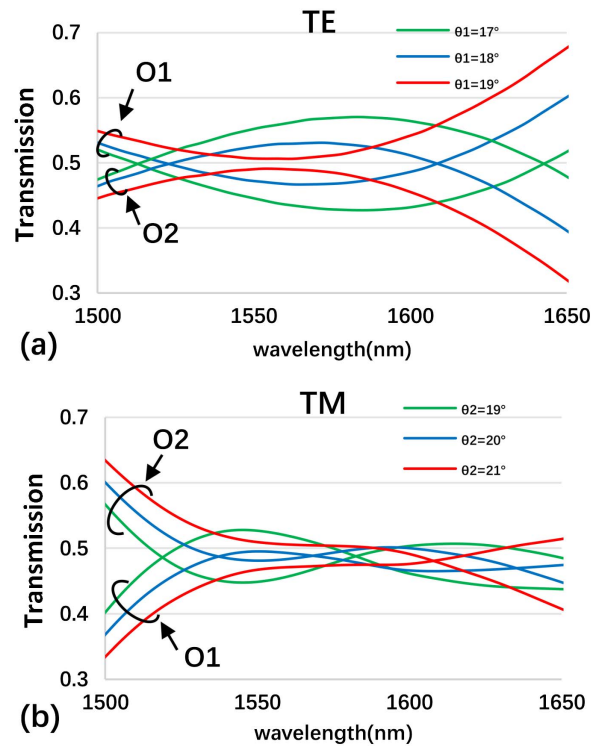


Fig. 3. Calculated transmittance spectra of the proposed splitter with different angle lengths: (a) TE mode and (b) TM mode. The parameters are $g_1 = 110 \text{ nm}$, $g_2 = 350 \text{ nm}$, $W = 500 \text{ nm}$, $R_1 = 35 \mu\text{m}$, $R_2 = 20 \mu\text{m}$, $\theta_1 = 18^\circ$, $\theta_2 = 20^\circ$, and $L = 5 \mu\text{m}$.

length of the straight $L = 5 \mu\text{m}$, so that the R_5 can be larger than $20 \mu\text{m}$, and the whole structure can still be kept compact.

The fabrication processes were performed based on a 220 nm thick commercial SOI wafer. An electron-beam lithography (Raith150 II) was carried out to define the pattern. After that, inductively coupled plasma dry-etching was applied to etch through the silicon core layer. Finally, a SiO_2 upper-cladding was deposited onto the wafer with a plasma-enhanced chemical vapor deposition process. Figure 4(a) shows the microscopic image of the fabricated 2×2 3 dB power splitter. Two identical 3 dB splitters connected with TE- or TM-type grating couplers (GCs) were fabricated on the same chip in order to measure the transmission spectra of the two polarization states simultaneously. Figure 4(b) shows the scanning electron microscope (SEM) picture of the coupling region. Figures 4(c) and 4(d) show the shallow etched TE-type GC and the fully etched TM-type GC. Figures 4(e) and 4(f) show the enlarged SEM picture of the coupling region for DC#1 and DC#2. In addition, we fabricated reference straight waveguides with TE/TM type GCs on the same chip for normalization. A broadband amplified spontaneous emission (ASE) light source and an optical spectrum analyzer were used to characterize the device. A fiber-type polarizer and polarization controller were used to confirm the polarization state of the input light.

First, we measured the spectra of the reference straight waveguide and the 3 dB power splitter, respectively. Then, we subtracted the spectra of the 3 dB power splitter from that of a straight waveguide to obtain the normalized transmittance. Figure 5 shows the measured normalized transmission spectra

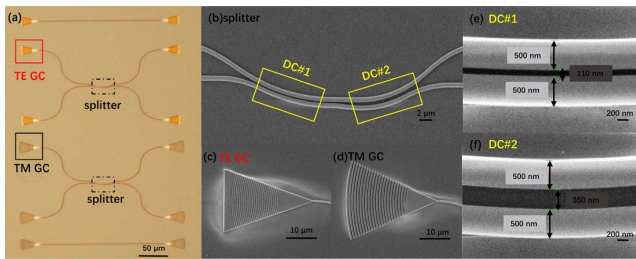


Fig. 4. (a) Microscopic image of the fabricated 2×2 3 dB power splitter. (b) SEM pictures of the fabricated 2×2 3 dB power splitter. (c), (d) SEM pictures of the GCs. (e), (f) Enlarged view of the coupling regions.

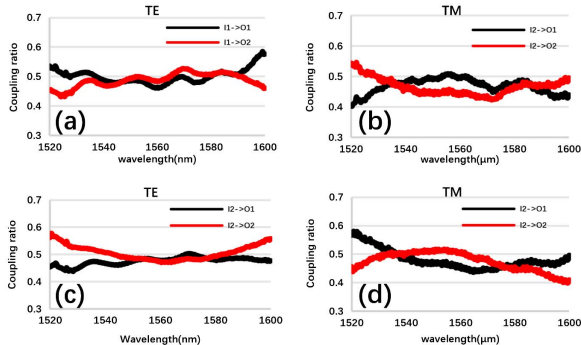


Fig. 5. Normalized transmittance spectra for the fabricated 2×2 3 dB power splitter based on bent DCs: (a) TE polarization from input1, (b) TM polarization from input1, (c) TE polarization from input2, and (d) TM polarization from input2.

of the fabricated 3 dB splitters for both the TE and TM polarizations. Due to the limitation of the light source, we only obtained the spectra within the range of the 1520–1600 nm. The deviation from the 3 dB level in the normalized transmission spectra is within ± 1 dB for the whole spectrum range. The splitting ratio is quite good, since the responses of the two input ports (I1, I2) are almost the same. The measured results agree well with the simulation for both polarizations. The excess losses for both polarizations are less than 1 dB. One should also note that excess loss < 0.5 dB is difficult to be measured due to the variation in normalization. There is maybe an over-estimation around 1520 nm, since it is the bandwidth edge of the GCs. More accurate results on the excess loss can be characterized by measuring the output power after passing through different numbers of splitters [13].

In order to check the fabrication tolerance of the present 3 dB power splitter, we have fabricated several splitters with different waveguide widths, i.e., $W = 490, 500$, and 510 nm. Figure 6 shows the measured normalized power splitting ratio of the TE and TM modes when light was injected into Port 1. One can find that the 3 dB power splitter can still work properly at a band from 1520 to 1600 nm when the width deviation is about 10 nm, which can be easily achieved by the modern nano-fabrication technologies [14].

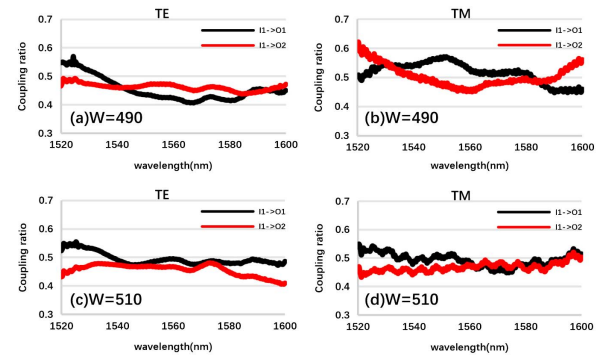


Fig. 6. Normalized transmittance spectra for the fabricated 2×2 3 dB power splitter with different waveguide widths: (a) TE polarization when $W = 490$ nm, (b) TM polarization when $W = 490$ nm, (c) TE polarization when $W = 510$ nm, and (d) TM polarization when $W = 510$ nm.

In conclusion, we have proposed and demonstrated a broadband and polarization-independent 2×2 3 dB power splitters based on bent DCs. The measurement results show that the proposed splitters have a splitting ratio varying from 45%:55% to 50%:50% in the wavelength band from 1520 to 1630 nm for both TE and TM polarizations, with an excess loss less than 1 dB. Furthermore, the total device length is less than 50 μm . Therefore, we believe the proposed power splitter would find its applications in numerous silicon photonic devices.

Funding. National Key Research and Development Plan of China (2016YFB0402502); National Natural Science Foundation of China (NSFC) (61377023, 61675178).

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